Water access and historical transport

Controls: Water Access Sum 1820; Railroads 1840, 1850 and 1861; Ocean and Great Lakes Access; Historical River and Canal Access; Waterway Reference Year.

1. The original water-access sum. Water Access Sum 1820 is one variable ranging from zero to three. Let I_{coast}, I_{river} and I_{canal} indicate whether the county or MSA intersects each original transport layer. Then

$$w_g^{1820} = I_{\text{coast}} + I_{\text{river}} + I_{\text{canal}}.$$

Each indicator is one when access is present and zero otherwise. A value of two means two types of access. The construction uses the project's original east-coast layer and the supplied buffered river and canal layers. It does not count routes, ports or traffic. Entering this sum assigns the same coefficient to an additional type of access, whichever type it is.

2. The separate access measures. Ocean and Great Lakes Access adds two indicators. Each equals one if any part of the county or MSA lies within 10 kilometres of the relevant modern shoreline. Historical River and Canal Access adds two further indicators for access within 10 kilometres of a river with documented steamboat operation and a canal operating in the selected year. Access is measured from the nearest part of the geographic unit, rather than its centre.

These four indicators receive separate coefficients. They differ from the original water-access sum in sources and construction. In particular, Great Lakes access is explicit here, and the original coast layer is not the modern ocean-shoreline layer. Selecting both sets adds both to the model; it does not replace one with the other.

3. The waterway year and railroads. The waterway year can be 1790 through 1860 in ten-year steps. A river or canal is active when the selected year falls between the recorded opening and closing dates, including both endpoints. This choice changes historical river and canal access and distance measures. It does not change the population instrument year, historical territory year, original 1820 water sum, modern shorelines or portage proxy.

Each railroad box adds a separate indicator equal to one if a mapped railroad intersects the county or MSA in that stated year. It measures network presence, rather than railroad length or distance to a station.

4. Missing values. Unknown operating dates can leave access uncertain. A known nearby active route establishes access; otherwise a potentially active nearby route can make the value missing. Missing selected controls exclude observations. The separate access measures cover the contiguous United States.